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Paraphrasing evades detectors of AI-generated text, but retrieval is an effective defense

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Abstract

The rise in malicious usage of large language models, such as fake content creation and academic plagiarism, has motivated the development of approaches that identify AI-generated text, including those based on watermarking or outlier detection. However, the robustness of these detection algorithms to *paraphrases* of AI-generated text remains unclear. To stress test these detectors, we build a 11B parameter paraphrase generation model (DIPPER) that can paraphrase paragraphs, condition on surrounding context, and control lexical diversity and content reordering. Using DIPPER to paraphrase text generated by three large language models (including GPT3.5-davinci-003) successfully evades several detectors, including watermarking, GPTZero, DetectGPT, and OpenAI's text classifier. For example, DIPPER drops detection accuracy of DetectGPT from 70.3% to 4.6% (at a constant false positive rate of 1%), without appreciably modifying the input semantics.

To increase the robustness of AI-generated text detection to paraphrase attacks, we introduce a simple defense that relies on *retrieving* semantically-similar generations and must be maintained by a language model API provider. Given a candidate text, our algorithm searches a database of sequences previously generated by the API, looking for sequences that match the candidate text within a certain threshold. We empirically verify our defense using a database of 15M generations from a fine-tuned T5-XXL model and find that it can detect 80% to 97% of paraphrased generations across different settings while only classifying 1% of human-written sequences as AI-generated. We open-source our models, code and data.[1]

1 Introduction

Large language models (LLMs) such as ChatGPT [Schulman et al., 2022] exhibit an unprecedented ability to write coherent and relevant long-form text in response to user-specified prompts. These abilities have sparked fears of malicious applications such as automatically generating fake news articles or homework answers [Stokel-Walker, 2022]. To defend against these use cases, several algorithms have recently been proposed to detect AI-generated text, including watermarking [Kirchenbauer et al., 2023a], GPTZero [Tian, 2023], DetectGPT [Mitchell et al., 2023], and OpenAI's text classifier [OpenAI, 2023a]. However, it remains unclear how robust these algorithms are to *paraphrase attacks*, in which AI-generated text from an LLM is rewritten by another (smaller) model to convey approximately[2] the same meaning but using different word choice and syntax.

[*]Work done as a PhD student at UMass, and partially as a student researcher in Google Research.
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[1]https://github.com/martiansideofthemoon/ai-detection-paraphrases
[2]We use the *quasi-paraphrase* definition of semantic equivalence [Bhagat and Hovy, 2013] in this paper.

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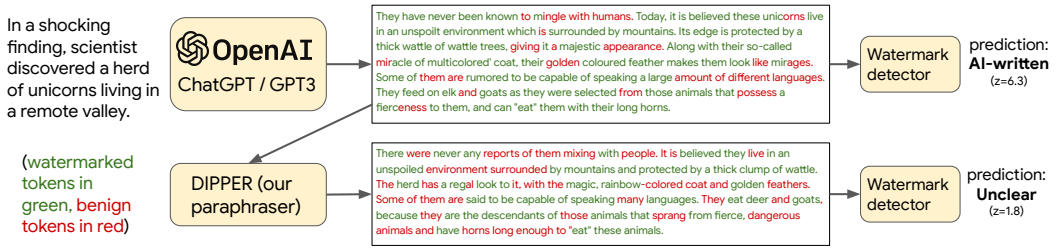

Figure 1: An overview of paraphrasing attacks with DIPPER on watermarked text [Kirchenbauer et al., 2023a]. The original model generation (top) contains several "green" watermarked tokens that are counted by a detector to judge whether the text was AI-generated. After paraphrasing, several green tokens are replaced with approximately semantically-equivalent red tokens, thereby fooling the detector. Actual outputs from a watermarked version of GPT2-XL and our paraphraser DIPPER.

In this paper, we first demonstrate the vulnerability of these existing detectors to paraphrase attacks (Section 3, 4). Such attacks require an *external* paraphraser model, since paraphrases generated by the base LLM are still susceptible to detection techniques such as watermarking. We train an 11B parameter paraphrase generation model called DIPPER (or **Di**scourse **P**ara**p**hras**er**) to execute these attacks. DIPPER possesses two unique features that help its outputs evade AI-generated text detectors: (1) **Paraphrasing long-form text in context:** Most modern paraphrasers are exclusively trained on sentence-level data, ignoring discourse-level information. However, many critical use cases of LLMs involve generating long-form text as responses to detailed user-specified prompts. Thus, we train DIPPER to paraphrase paragraph-length texts, re-order content, and optionally leverage context such as user prompts; (2) **Controlling output diversity:** Another weakness of existing paraphrasers is that they lack an easy way to control output diversity. An attacker may want to apply just the minimum amount of paraphrasing to evade a detector. DIPPER provides users with two intuitive scalar control knobs at inference time (lexical diversity, content reordering) that are trained end-to-end.

We use DIPPER to attack several recently proposed AI-generated text detectors (see Figure 1 for an attack overview). Experiments on multiple tasks and LLMs (including GPT3.5-davinci-003) show that after paraphrasing with DIPPER, a substantial fraction of AI-generated texts are misclassified as human-written texts by all detectors. For example, DetectGPT [Mitchell et al., 2023] correctly detects 70.3% of AI-generated sequences from GPT2-XL, but after paraphrasing, its detection rate drops to only 4.6%[3] despite minimal semantic modification. We confirm the validity of DIPPER's paraphrases through several automatic evaluations and a human evaluation of semantic similarity.

Given the vulnerability of AI-generated text detectors to paraphrasing, how can we defend against such attacks? In the second part of our paper (Section 5), we propose to use *retrieval* methods to detect AI-generated text instead of relying on statistical properties of the text or watermarking. First, an LLM API provider stores every output generated by their model in a database. The API provider then offers a service in which a semantic representation of a candidate text is compared to representations of every generation stored in the database. The search focuses on the *semantics* of the input and can leverage both standard IR methods such as BM-25 [Robertson et al., 1995] as well as semantic vector representations such as P-SP from Wieting et al. [2022]. Since paraphrasing does not modify the semantics of the input, this algorithm is robust to paraphrasing attacks. Specifically, we find that 97.3% of PG19 paraphrases and 80.4% of Wikipedia paraphrases are successfully detected in a large database of over 15M generations, at a 1.0% false positive rate. We extensively discuss the limitations and scalability of retrieval-based detection in Section 5.4.

In contrast to concurrent work that also uses paraphrasing to attack AI-generated text detectors [Sadasivan et al., 2023], our work offers more comprehensive attack experiments, a new and more powerful paraphraser, human evaluations of paraphrase quality, and finally a novel defense mechanism based on retrieval to combat such attacks. To spur future research in this area, we will release our DIPPER model, data, and a codebase for evaluating both existing detectors and our retrieval-based method.

[3]These detection rates were computed at a constant false positive rate (FPR) of 1%. Due to the importance of low FPR in this task, we recommend using a fixed low FPR rather than AUC-ROC values; see Section 4.1.

2 Background on detectors of AI-generated text

In this section, we provide a brief overview of existing algorithms for detecting AI-generated text detection (see Appendix E for a detailed version). We also contrast our work to Sadasivan et al. [2023], a concurrent effort which notes the efficacy of paraphrasing attacks but does not consider a retrieval-based defense in its pessimistic conclusion about the fate of AI-generated text detection.

A **watermark** is a modification to the generated text that can be detected post-hoc by an algorithm while remaining imperceptible to human readers. Effective watermarks are difficult to remove and have little effect on the quality of generated text. Prior work has watermarked natural language using syntax tree manipulations [Topkara et al., 2005, Meral et al., 2009], and this area has received renewed interest with the advent of LLMs [Abdelnabi and Fritz, 2021, Grinbaum and Adomaitis, 2022]. Most recently, Kirchenbauer et al. [2023a] proposed a simple algorithm that watermarks LLMs by slightly perturbing its probability distribution while generating text.

Statistical outlier detection methods detect AI-generated text based on its artifacts [See et al., 2019, Holtzman et al., 2020] instead of modifying the generative algorithm. Early methods detect statistical irregularities in entropy [Lavergne et al., 2008] and perplexity [Beresneva, 2016], while Gehrmann et al. [2019] introduced the GLTR visualizer to assist humans in detecting AI-generated text. The release of ChatGPT prompted the development of two new tools: closed-source GPTZero [Tian, 2023] and open-source DetectGPT [Mitchell et al., 2023]. The latter uses the observation that AI-generated text tends to have significantly higher LLM likelihood than meaningful perturbations of it.

Classifier methods train models to distinguish human-written text from AI-generated text. Early efforts used classifiers to detect fake reviews [Hovy, 2016] and news [Zellers et al., 2019], while others examined classifier performance across domains [Bakhtin et al., 2019] and decoding strategies [Ippolito et al., 2020]. Most recently, OpenAI fine-tuned a GPT model to perform this task and released it as a web interface [OpenAI, 2023a]. Their model uses generations from 34 LLMs, with the human-written text from Wikipedia, WebText, and their internal human demonstration data.

Comparison to Sadasivan et al. (2023): In recent concurrent work, Sadasivan et al. [2023] also demonstrate the utility of paraphrasing attacks against AI-generated text detectors. Our experiments encompass more tasks, detection algorithms, and larger LMs like GPT3.5. Additionally, we propose a discourse-level paraphrase model (DIPPER) that is much more suited to long-form text than the off-the-shelf sentence-level paraphrasers used in their paper. More importantly, our retrieval-based defense *directly contradicts* the "impossibility result" of Sadasivan et al. [2023] and its associated proof, which states that an optimal detector will perform at random as the quality of LLM-generated text approaches that of human-written text. The quality of generated text is irrelevant to our detector's accuracy because it relies only on a corpus search, and thus the proof is inapplicable. Other concurrent work [Chakraborty et al., 2023] has also shown the proof's invalidity in practical settings.

3 Building a controllable discourse paraphraser

Having outlined existing methods to detect AI-generated text, we now focus on a simple attack against all detection techniques: *paraphrasing* the generated text. Intuitively, paraphrasing alters the statistical properties of AI-generated text, which can fool outlier detection or classifiers while also reducing the number of watermarked tokens (Figure 1). To evade such detectors, a paraphraser must be able to handle *context* in the form of prompts or multi-sentence inputs. Its behavior should also be *controllable* in order to make as many/few changes as needed to evade a given detector. In all cases, it should not appreciably change the input semantics. Finally, to evade watermarking, the paraphraser must be different from the watermarked model, as otherwise the paraphrases will also be watermarked. Below, we detail how we construct a paraphraser (DIPPER) with all these properties.[4]

Constructing paraphrase data: Our process involves fine-tuning a LLM on a parallel dataset of paragraph-level paraphrases, which we modify to model control, external context and content reordering. We leverage the PAR3 dataset [Thai et al., 2022], which contains multiple translations of non-English novels into English aligned at a paragraph level, which we treat as paraphrases. More formally, let p and q be aligned paragraphs, where $p_1, p_2 ... p_N$ denote sentences of p and $q_1, q_2, ... q_M$

[4]To better ground DIPPER's abilities in prior work, we survey existing paraphrase models in Appendix D.1.